- $Q(A, B) = 3.25$
- $Q(B, C) = 4.25$
- $Q(C, E) = 3.75$
- $Q(E, A) = 2.75$

All other existing edges stay at 5:

- $Q(A, D) = 5,\ Q(B, D) = 5,\ Q(C, D) = 5,\ Q(D, C) = 5$

2️⃣ Linear Neural Network that Represents This Q-table

Use a **one-layer linear network**:

- **Input layer:** one-hot state vector $[A, B, C, D, E]$
- **Output layer:** one neuron per "action = next state" $[A, B, C, D, E]$

Weights form a 5×5 matrix W.
For state s and action "go to next state a", the Q-value is:

$$Q_\theta(s, a) = W_{s,a}$$

Forward pass for chosen (s,a):

- Input is one-hot at row s
- Read the output neuron for column a
- That output equals the single weight $W_{s,a}$

So after learning, the **weight matrix is**:

from \ to	A	B	C	D	E
A	5	**3.25**	5	**5**	5
B	5	5	**4.25**	**5**	5
C	5	5	5	**5**	**3.75**
D	5	5	**5**	5	5
E	**2.75**	5	5	5	5

(Entries that don't correspond to real edges are never used; they can stay at 5.)